



Tiny Splatoon

不侵權資產製作的斯普拉遁

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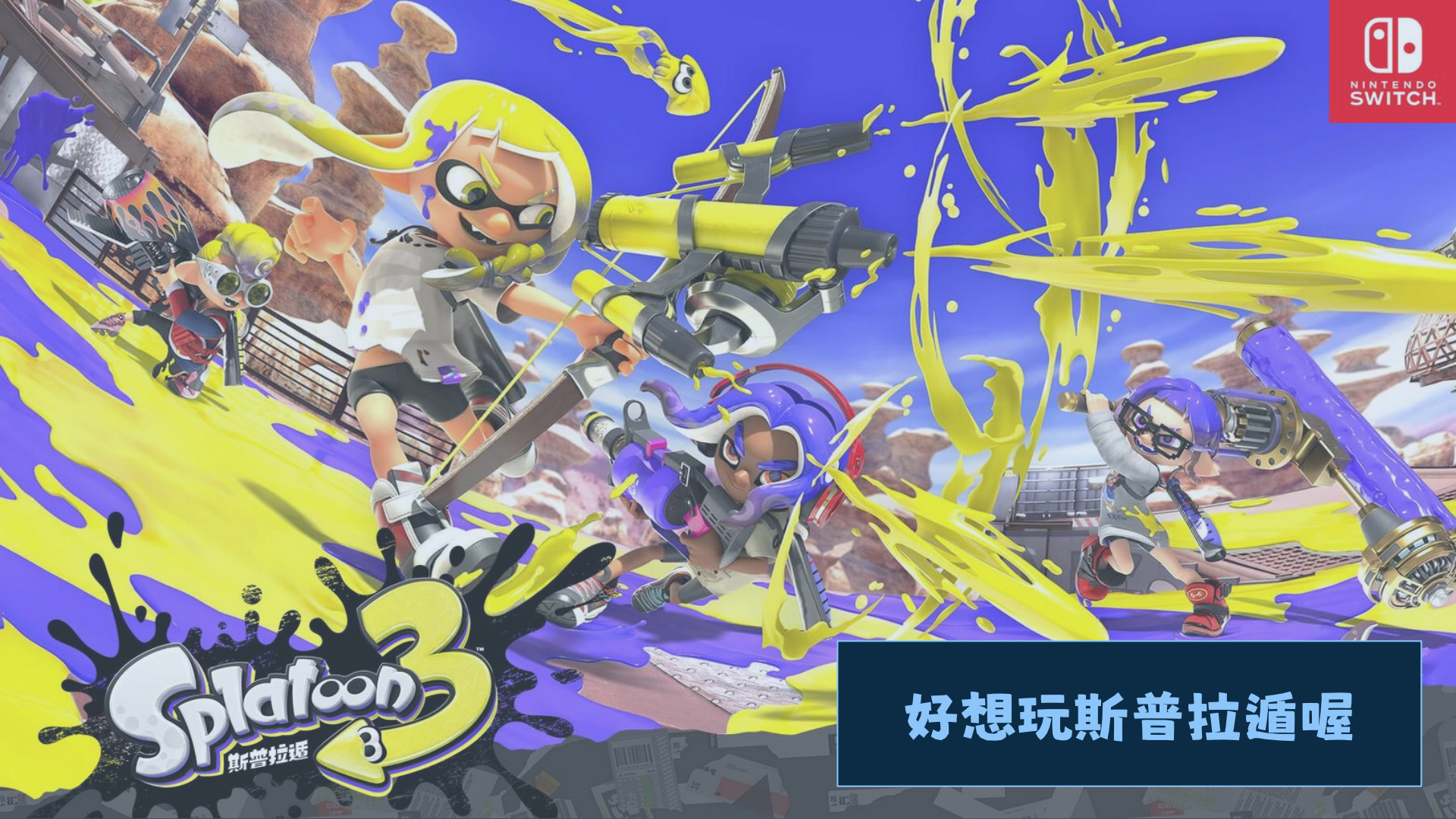
心得

The background is a dark blue gradient. Scattered throughout are several translucent blue bubbles of varying sizes, some with white highlights to give them a 3D effect. In the center, there is a white circle containing the number '01' in a white, italicized sans-serif font.

01

About the Project

實作內容及想法



Splatoon
斯普拉遁
3

好想玩斯普拉遁喔

塗抹、被塗抹、塗抹回去 使用墨汁的占地對戰



以隊伍形式塗抹地面，
塗抹面積較多的隊伍為勝利！

Motivation

- 只能玩別人的斯普拉遁 🥲
- NSO 還要收錢
- 自己做一款來玩
- 希望不會被最強法務部關心
- 小目標：做到受不了想手搓遊戲引擎的程度

Features

- Localhost / Online
- Camera Shaking
- Audio
- Selectable Weapons
- Keyboard & mouse

Weapons



Shooter

- 水槍 / 銀噴
- 中距離



Brush

- 筆刷 / 拖把
- 近距離



Slosher

- 潑水桶 / 洗衣機
- 中遠距離

The background is a dark blue gradient. Scattered across the frame are several translucent, 3D-rendered bubbles of varying sizes. Some bubbles show highlights and shadows, giving them a realistic, floating appearance. They are positioned in the corners and around the central text.

02

System Architecture

遊戲系統設計

Module Layer

Audio, Input, GUI Mgr

Scene Mgr

Network Mgr

Game

Lobby

GameWorld

Login

Players

Component Sys.

Splat Mechanics

(大概的)

Modules (Detail)



https://github.com/SoWiEee/Tiny-Splatoon/blob/main/README_detailed.md

03

Trouble-Shooting

實作問題與改善方案

Problems

- Memory, Double Free
- $\Rightarrow$ Smart Pointer
- Circular Include
- $\Rightarrow$ Forward Declaration
- God Object
- $\Rightarrow$ Let it go :)

Smart Pointer (C++11)

- 還在手動 new/delete 嗎？
- 自動管理記憶體生命週期，解決 memory leak
- 獨佔指標 `unique_ptr<T>`
 - 當指標被銷毀時，物件也會自動被刪除
- 共享指標 `shared_ptr<T>`
 - 只有當最後一個擁有者消失時，物件才會被刪除

The background is a dark blue gradient. Scattered throughout are several translucent, light blue bubbles of varying sizes. Some bubbles have a white highlight on their upper-left edge, giving them a 3D effect. They are positioned in the corners and around the central text.

04

Reflection

實作心得

Conclusion

- 會寫 C++ 了
- 圖學 Debug
- 架構好難
- 效能 & 效率

The background is a dark blue gradient. Scattered throughout are several translucent, 3D-rendered bubbles of varying sizes. Some bubbles have bright highlights, giving them a glossy appearance. They are positioned in the corners and around the central text, creating a clean, modern aesthetic.

05

Env. & Ref.

開發環境與參考資料

Development Environment

- C++17 in Windows 11 25H2
- Visual Studio 2022 Release (amd64) 17.14.19 Compiler
- OpenGL 4.5 (core), GLAD, GLFW 3.4 (x64)
- GLM 1.0.2
- ImGui 1.92.5
- GameNetworkingSockets
- MiniAudio

Reference

- Splatoon 3
- Lobby BGM – Life Will Change (instrumental)
- Game BGM – It's Going Down Now (Instrumental)
- 音效資源、死亡音效
- Tunneling Tool – Playit.gg

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Thanks For Your Attention

Contribute to This Repo

<https://github.com/SoWiEee/Tiny-Splatton>

Or give me a  !

Download & Play

1. Download Game at <https://reurl.cc/qK25YD>
2. Unzip and run [build/Tiny-Splatton.exe](#)
3. Connect to womany-academic.gl.at.ply.gg:38439



Mint_Ice ZTMY 上午 12:34

你的下載zip到解壓縮完大概多久



糟糕先生 不要四層 上午 12:34

5分鐘左右吧 (已編輯)

114 學年度第一學期 演算法 第三次作業

1. Problem: Huffman Coding 壓縮

給定一個 PGM image 檔案，請利用 Huffman Coding 演算法設計與實作檔案壓縮程式：

- i. Input: xxx.pgm (using IrfanView (<https://www.irfanview.com/>) to covert jpeg, gif, and more into pgm ASCII format) ([PGM P2](#))
- ii. Output: a compressed file xxx.hc
- iii. Functions:

謝謝壓縮演算法

- a. 壓縮(Compression) <= 須說明如何將 Huffman Tree and Codes 置於檔案中
- b. 解壓縮(Uncompression)

Game Rules

- 會自動分配紅隊/綠隊
- 鍵盤 WASD 移動、左鍵普通攻擊、空白鍵跳躍
- 按住 Shift 潛入墨水、R 投擲炸彈、Q 啟動火箭筒（3發/6秒）
- 想辦法通靈回血機制（簡單來說會自然回血）
- 重生時間 4 秒、遊戲時間 3 分鐘
- 根據塗地面積結算 Win/Lose